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1. Introduction

The **Multi-Scale Spatiotemporal Graph Mamba (MSGM)** model was selected because it uses the FACED dataset, performs subject-independent binary emotion classification, represents EEG channels as a meaningful graph and reports strong performance. On FACED, MSGM achieved **$63.17 \pm 3.62\%$ accuracy** and **$76.01 \pm 3.74\%$ reported F1 score**. BiDANN achieved slightly higher accuracy at **63.36%**, while MSGM achieved the highest reported F1 score among the methods compared.

2. Why MSGM Was Selected

Selection criterion	Justification
Dataset match	Both studies use FACED
Task match	Both use binary positive-versus-negative emotion classification
Track suitability	EEG channels naturally form graph nodes and channel relationships form edges
Strong literature result	MSGM reported 63.17% accuracy and 76.01% F1
Architecture relevance	It combines frequency, spatial and temporal EEG information
Reproduction value	It provides a substantial graph-based literature benchmark rather than a simple classifier

3. MSGM Architecture

Preprocessed EEG → 20-second overlapping segments → Multiple second-level temporal windows → Seven-band rPSD extraction → Multi-scale feature tensors → Global and local EEG graphs → Shallow and deep GCN branches → Spatiotemporal feature fusion → MSST-Mamba → Unified embedding → Binary classifier

3.1 Multi-scale temporal segmentation

The raw EEG recording is first divided using:

- first-level window: **20 seconds**;
- first-level hop: **4 seconds**;
- overlap: **16 seconds**.

Each 20-second segment is then divided into several smaller temporal windows. Shorter windows capture rapid EEG changes, while longer windows capture more sustained patterns. The paper explains this two-level segmentation procedure but does not clearly list the exact second-level window lengths and hop values.

3.2 rPSD feature extraction

For every EEG channel and temporal sub-segment, relative power spectral density is calculated using **Welch’s method**.

Seven frequency bands are used:

Frequency band	Range
Delta	1–4 Hz
Theta	4–8 Hz
Alpha	8–12 Hz
Low beta	12–16 Hz
Beta	16–20 Hz
High beta	20–28 Hz
Gamma	30–45 Hz

rPSD represents the proportion of total selected EEG power belonging to a particular band. These seven rPSD values become the features of each EEG-channel node.

3.3 Multi-scale feature tensors

For scale (k), MSGM stores the extracted features as: $F_k \in \mathbb{R}^b \times n_k \times c \times f$

where:

- (b) = batch size;
- (n_k) = number of temporal sub-segments at scale (k);
- (c) = number of EEG channels;
- (f) = number of rPSD features.

Example using the reproduced configuration

For FACED:

- batch size = 32;
- channels = 32;
- features = 7;
- sequence lengths = 16, 23, 36 and 77.

Therefore, the four inputs are:

Scale	Tensor shape
Scale 1	(32 × 16 × 32 × 7)
Scale 2	(32 × 23 × 32 × 7)
Scale 3	(32 × 36 × 32 × 7)
Scale 4	(32 × 77 × 32 × 7)

For example: $32 \times 16 \times 32 \times 7$ means:

- 32 EEG samples in the batch;
- 16 temporal sub-segments per sample;

- 32 EEG channels per sub-segment;
- 7 rPSD values per channel.

A value such as: $F[5, 10, 12, 2]$

represents the third frequency feature of the thirteenth channel, in the eleventh temporal segment, of the sixth batch sample.

3.4 Graph construction

Each EEG channel becomes **one graph node**. MSGM constructs **two graphs**.

Graph	Function
Global graph	Models relationships between channels across the complete scalp
Local graph	Keeps only relationships between channels in the same predefined region

The global graph uses:

- Pearson correlation to measure similar channel behaviour;
- Manhattan distance to measure feature difference;
- Gaussian-kernel weighting to assign edge strength.

The local graph retains a global edge only when both channels belong to the same predefined scalp region.

Graph calculation example

Consider one scale: $F \in \mathbb{R}^{32 \times 16 \times 32 \times 7}$

First, the paper averages over the batch dimension: $32 \times 16 \times 32 \times 7 \rightarrow 16 \times 32 \times 7$

The temporal and feature dimensions are then combined:

$$16 \times 32 \times 7 \rightarrow 32 \times (16 \times 7) \rightarrow 32 \times 112$$

The 32 rows now represent the 32 EEG channels, while each channel has a 112-dimensional vector. Pairwise channel relationships generate a: 32×32 adjacency matrix.

3.5 Shallow and deep GCN branches

MSGM uses four graph branches:

1. shallow global GCN;
2. deep global GCN;
3. shallow local GCN;
4. deep local GCN.

The shallow GCN captures direct neighbouring relationships. The deeper GCN captures multi-hop relationships.

A linear projection also creates a base embedding directly from the input features. The model combines: Base feature embedding + Shallow GCN output + Deep GCN output separately for the global and local branches.

3.6 MSST-Mamba

After graph processing, Mamba learns temporal relationships. The processing order is:

RMSNorm → Input Linear projection → Split into main and gate paths → Conv1D on main path → SiLU activation → Selective State-Space Model → Multiply by activated gate → Output Linear projection → Residual addition

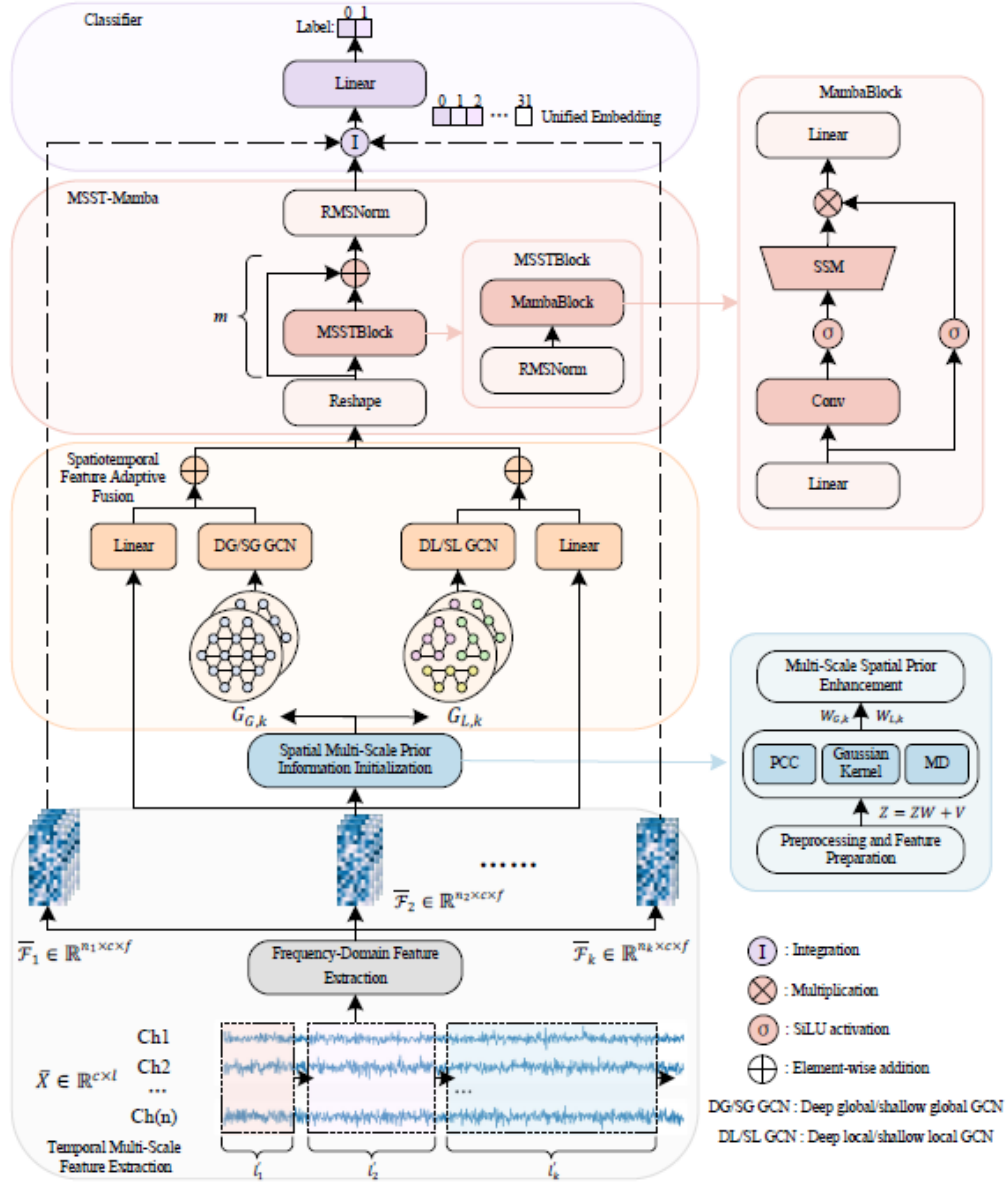


Fig.1. MSGM Architecture

4. Original Paper Results on FACED

Model	Accuracy, mean $\pm$ std	Reported F1, mean $\pm$ std
BiDANN	63.36 $\pm$ 7.01%	73.82 $\pm$ 6.36%
EmT	60.80 $\pm$ 6.50%	74.00 $\pm$ 5.80%
MSGM	63.17 $\pm$ 3.62%	76.01 $\pm$ 3.74%

5. Reproduction Scope

Component	Reproduction target
Dataset	FACED
Classification	Binary negative versus positive
Neutral class	Excluded
Node	One EEG channel
Node features	Seven-band rPSD
Temporal processing	Multiple temporal scales
Graphs	Global and local
GCN branches	Shallow and deep
Temporal model	One MSST-Mamba layer
Evaluation	Subject-independent cross-validation
Main comparison	Published MSGM FACED accuracy

6. Exact Reproduction Requirements and Availability

Requirement	Availability	Reproduction decision
FACED processed subject files	Available	All 123 available subject files were used
EEG shape and channel count	Available	28 trials, 32 channels and 7,500 samples per trial
Sampling rate	Available	250 Hz
Emotion metadata	Available	Used to construct binary labels
Exact binary mapping used by MSGM	Partially specified	Negative and positive categories were mapped explicitly; neutral was excluded
First-level segmentation	Available	20-second window and 4-second hop
Exact second-level windows	Unavailable	Replaced with 4, 3, 2 and 1 seconds

Requirement	Availability	Reproduction decision
Exact second-level hops	Unavailable	Replaced with 75% overlap
Frequency bands	Available	Same seven bands were used
Exact Welch parameters	Not fully reported	Notebook implementation was used and documented
Exact local-region mapping source	Not fully reported	Available MSGM implementation was followed
Exact test-subject identities	Unavailable	Deterministic subject-wise folds were created
Original random seed	Unavailable	Seed 42 was used
Original trained checkpoint	Unavailable	The model was trained from initialization
Original F1 averaging type	Unavailable	Binary, macro and weighted F1 were all reported
FACED-specific published parameter count	Unavailable	Reproduced count was reported separately

7. Notebook Design

Two notebooks were used.

7.1 Main Reproduction Notebook

Environment setup → Dataset discovery and validation → Metadata and label construction → Multi-scale rPSD extraction → Feature caching → Subject-wise fold generation → DataLoader validation → MSGM smoke test → Fold training and partial saving

7.2 Resume notebook

Restore cached features → Restore completed fold results → Audit checkpoints → Delete incomplete checkpoint only → Resume remaining folds → Combine ten-fold results → Generate metric and runtime summaries → Generate pooled predictions

The resume notebook was necessary because complete cross-validation required multiple long-running training sessions. It prevented completed folds from being retrained after a session interruption.

8. Experimental Configuration

Setting	Value used	Source/status	Explanation
Dataset	FACED	Original paper and reproduction	Same dataset used in both
Subjects	123	FACED dataset / reproduction	All available subjects were validated

Setting	Value used	Source/status	Explanation
Trials per subject	28	FACED dataset	Original stored trial count
EEG channels	32	FACED dataset and original paper	Same FACED channel count
Samples per trial	7,500	Reproduction dataset files	Confirmed from loaded arrays
Sampling rate	250 Hz	Original paper and FACED	Same value used
Task	Binary classification	Original paper and reproduction	Negative versus positive
Neutral handling	Excluded	Reproduction decision	Implemented explicitly in notebook
First-level window	20 seconds	Original paper	Reproduced directly
First-level hop	4 seconds	Original paper	Reproduced directly
Second-level windows	4, 3, 2 and 1 seconds	Reproduction replacement	Exact paper values were unavailable
Second-level overlap	75%	Reproduction replacement	Exact paper hop settings were unavailable
Sequence lengths	16, 23, 36 and 77	Reproduction output	Generated from the replacement window settings
Frequency bands	7	Original paper	Reproduced directly
Batch size	32	Original paper	Reproduced directly
Hidden dimension	32	Original paper	Reproduced directly
GCN depths	1 and 2	Original paper	Reproduced directly
Chebyshev order	4	Available implementation	Not clearly reported as a separate table value in the paper
MSST-Mamba layers	1	Original paper	Reproduced directly
Mamba state dimension	16	Original paper	Reproduced directly
Mamba convolution kernel	4	Original paper	Reproduced directly
Mamba expansion factor	2	Available implementation	Implementation setting
Dropout	0.25	Original paper	Reproduced directly
Loss	Cross-entropy	Original paper	Reproduced directly
Label smoothing	0.1	Original paper	Reproduced directly

Setting	Value used	Source/status	Explanation
Optimizer	AdamW	Original paper	Reproduced directly
Learning rate	(3×10^{-4})	Original paper	Reproduced directly
Weight decay	(10^{-2})	Reproduction implementation	Explicit optimizer setting in notebook
Maximum epochs	30	Original FACED setting	Reproduced directly
Early-stopping patience	5	Original paper	Reproduced directly
Random seed	42	Reproduction decision	Original seed unavailable
Test folds	10	Reproduction protocol	Created to test 12 subjects per fold
Test subjects per fold	12	Original principle and reproduction	Same number as original protocol
Hardware	Tesla T4	Reproduction environment	Original paper used RTX 3070Ti
Trainable parameters	221,480	Reproduction output	Count produced by the 32-channel FACED model

8.1 Why the parameter count is different

The original paper reports **349,218 parameters**, but this number is given in the section describing deployment on the **62-channel SEED dataset**, not as a separate FACED-specific parameter count.

9. Evaluation Metrics

Only metrics actually calculated in the notebooks are reported.

Metric	Meaning	Why required
Accuracy	Percentage of all predictions that are correct	Enables direct comparison with published MSGM accuracy
Binary F1	Harmonic mean of positive-class precision and recall	Measures positive-class balance
Macro-F1	Mean of negative- and positive-class F1	Gives equal importance to both classes
Weighted F1	F1 weighted by class support	Provides support-aware overall performance
Positive precision	Correct positive predictions divided by all predicted positives	Measures reliability of positive predictions
Positive recall	Correct positive predictions divided by all actual positives	Measures how many positive samples are detected

Metric	Meaning	Why required
Per-class metrics	Precision, recall and F1 separately for each class	Shows whether one class is easier
Confusion matrix	Counts of each correct and incorrect outcome	Shows the exact direction of errors
Mean $\pm$ standard deviation	Average and variability across folds	Measures cross-subject stability

10. Reproduced Results

10.1 Ten-fold subject-independent results

Metric	Mean $\pm$ standard deviation	Minimum	Maximum
Accuracy	57.52 $\pm$ 2.91%	52.20%	60.30%
Binary F1	53.68 $\pm$ 13.27%	17.23%	64.60%
Macro-F1	56.09 $\pm$ 5.48%	41.81%	60.17%
Weighted F1	56.09 $\pm$ 5.48%	41.81%	60.17%
Positive precision	59.33 $\pm$ 4.08%	54.05%	64.29%
Positive recall	53.15 $\pm$ 17.72%	9.95%	72.45%

10.2 Pooled confusion matrix

Actual class	Predicted negative	Predicted positive
Negative	2,674	1,646
Positive	2,024	2,296

11. Original Versus Reproduced Results

Measure	Original MSGM	Reproduced MSGM	Difference	Comparison status
Accuracy	63.17 $\pm$ 3.62%	57.52 $\pm$ 2.91%	−5.65 percentage points	Directly comparable
Reported F1	76.01 $\pm$ 3.74%	—	—	Original F1 type unspecified
Binary F1	Not reported separately	53.68 $\pm$ 13.27%	—	Descriptive only
Macro-F1	Not reported separately	56.09 $\pm$ 5.48%	—	Descriptive only
Weighted F1	Not reported separately	56.09 $\pm$ 5.48%	—	Descriptive only

Measure	Original MSGM	Reproduced MSGM	Difference	Comparison status
Channels	32 for FACED	32	Same	Comparable
Test subjects per fold	12	12	Same number	Exact IDs differ
Hardware	RTX 3070Ti	Tesla T4	Different	Runtime not directly comparable
Parameters	No FACED-specific count reported	221,480	Unknown	No valid direct comparison

12. Source Differences

Difference	Original information	Reproduction decision	Possible effect
Fold identities	Exact subject IDs unavailable	Deterministic subject-wise folds	Different unseen subjects may change difficulty
Second-level windows	Exact values unavailable	4, 3, 2 and 1 seconds	Different temporal representation
Second-level hops	Exact values unavailable	75% overlap	Different number of sub-segments
Random seed	Not reported	Seed 42	Different initialization
Hardware	RTX 3070Ti	Tesla T4	Different runtime and numerical environment
Trained weights	Unavailable	Trained from initialization	Different learned solution
F1 definition	Unspecified	Three F1 versions reported	Exact F1 comparison impossible
FACED parameter count	Not reported	221,480 measured	No parameter-identical confirmation

13. Runtime

Runtime measure	Reproduced value
Mean completed epochs per fold	13.50
Total training time	20.27 hours
Total validation time	5.91 minutes
Total test time	25.12 seconds
Complete ten-fold runtime	20.39 hours

Runtime measure	Reproduced value
Mean test time per fold	2.51 seconds

Early stopping reduced the average number of completed epochs from the maximum of 30 to 13.5 epochs.

Conclusion

MSGM was selected as a strong literature-based benchmark because it used the same FACED dataset, the same binary emotion-classification setting and a meaningful EEG graph architecture. The reproduction retained the main MSGM components: multi-scale rPSD features, global and local EEG graphs, shallow and deep GCN branches, MSST-Mamba temporal modeling and subject-independent evaluation.

The reproduced model achieved:

- **$57.52 \pm 2.91\%$ accuracy;**
- **$56.09 \pm 5.48\%$ macro-F1;**
- **$53.68 \pm 13.27\%$ binary F1.**

The reproduced accuracy was **5.65 percentage points below** the original MSGM accuracy of 63.17%. Exact F1 replication could not be assessed because the original paper did not define its F1 averaging method.